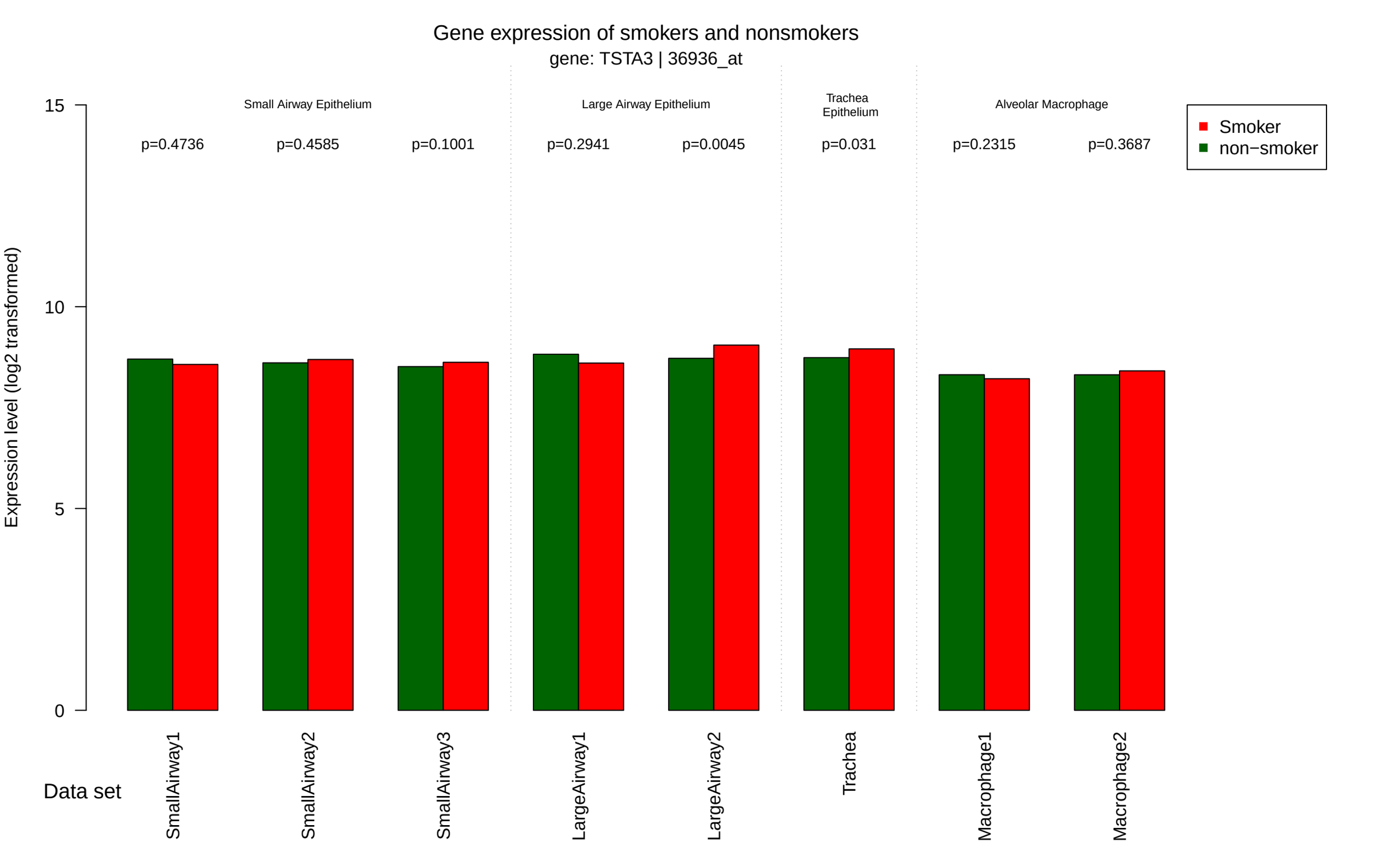
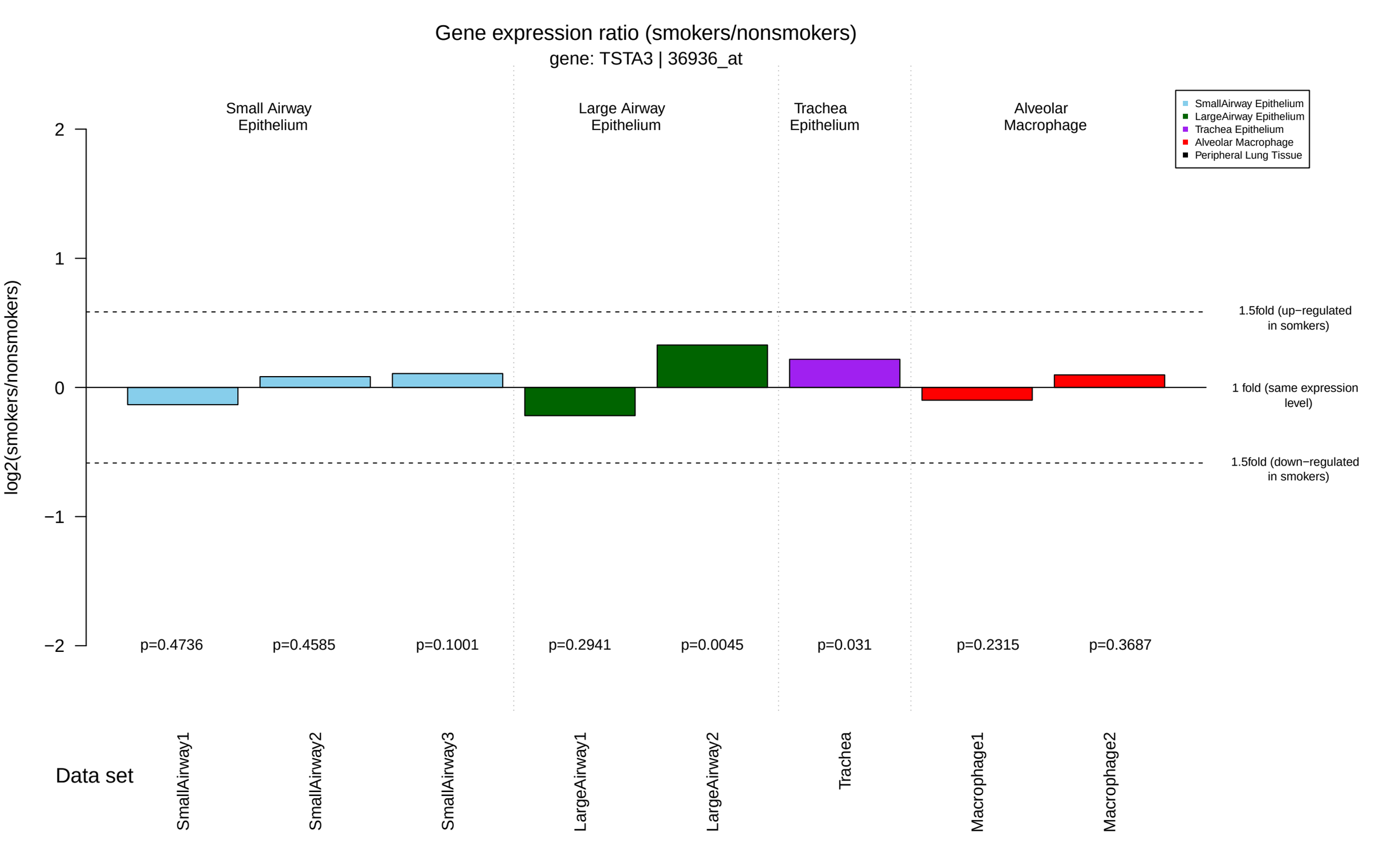
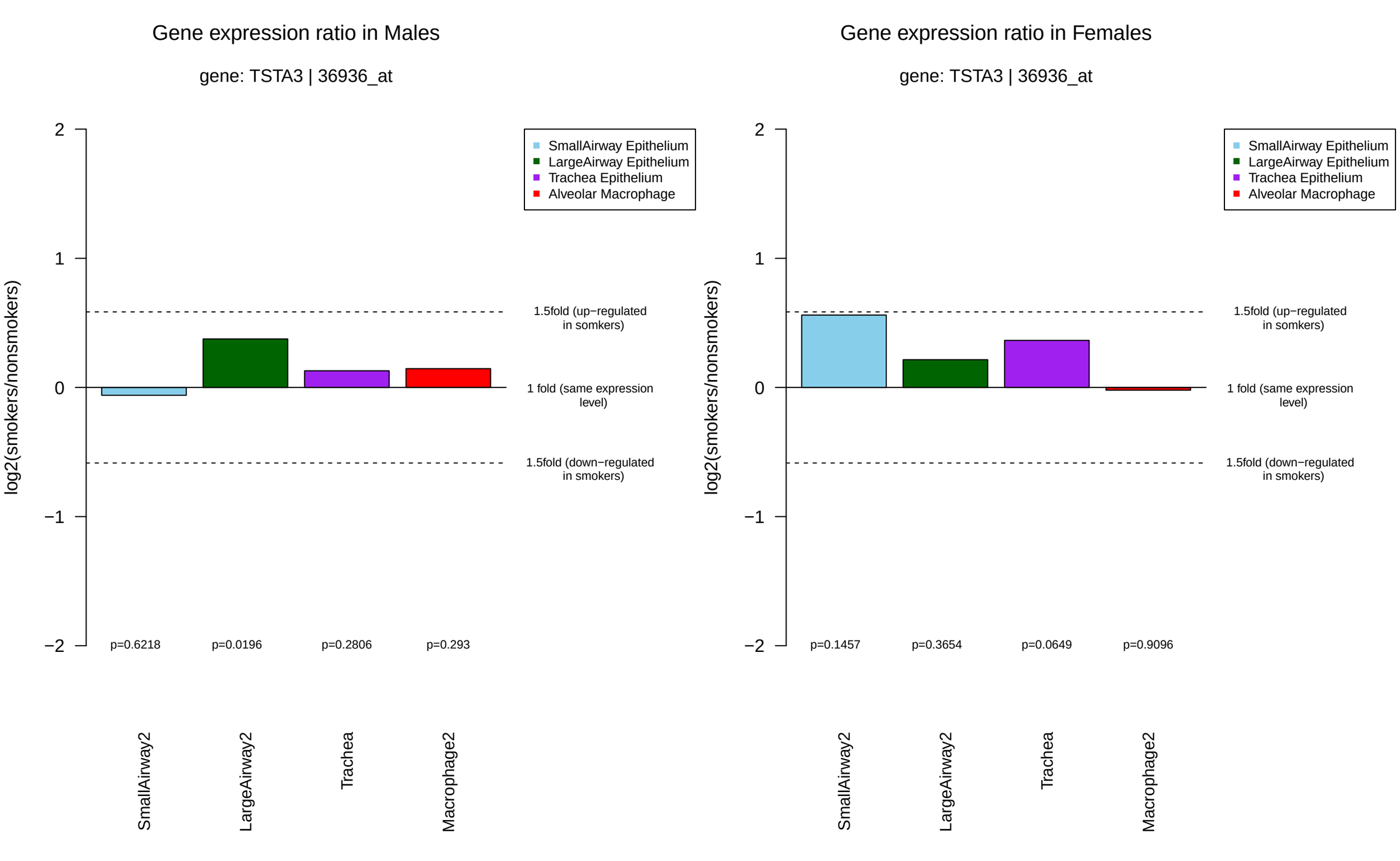
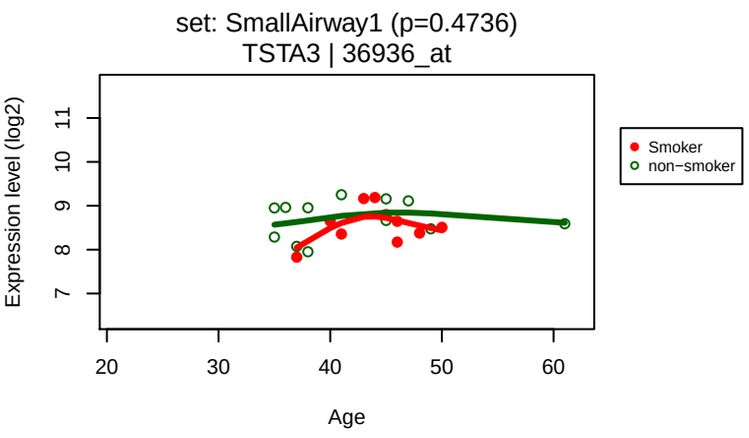
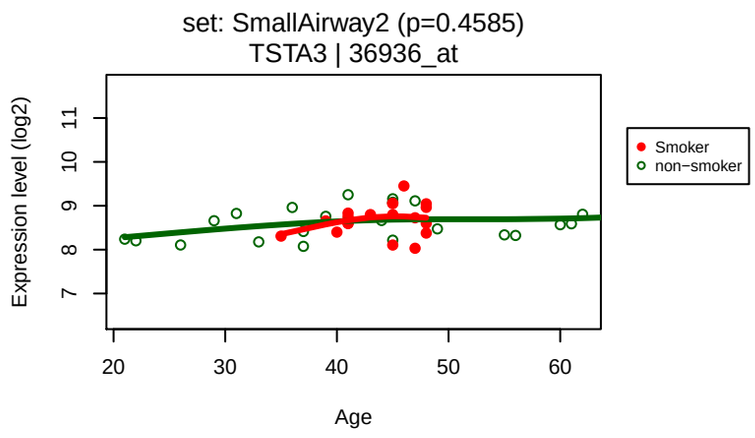
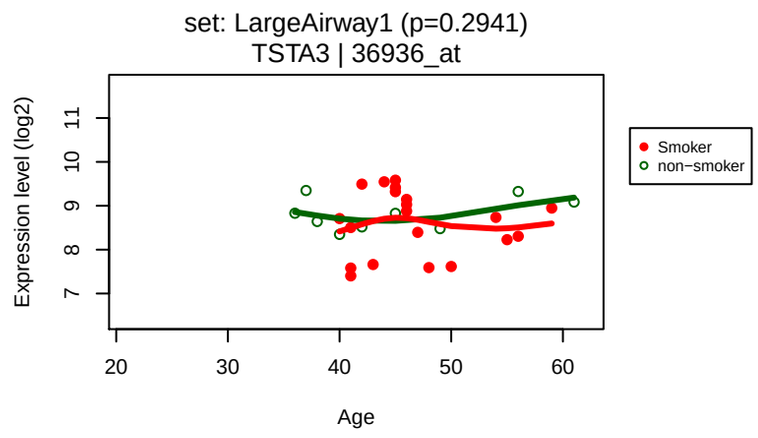

Gene expression of smokers and nonsmokers
gene: TSTA3 | 36936_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4736
p=0.4585
p=0.1001
p=0.2941
p=0.0045
p=0.031
p=0.2315
p=0.3687
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

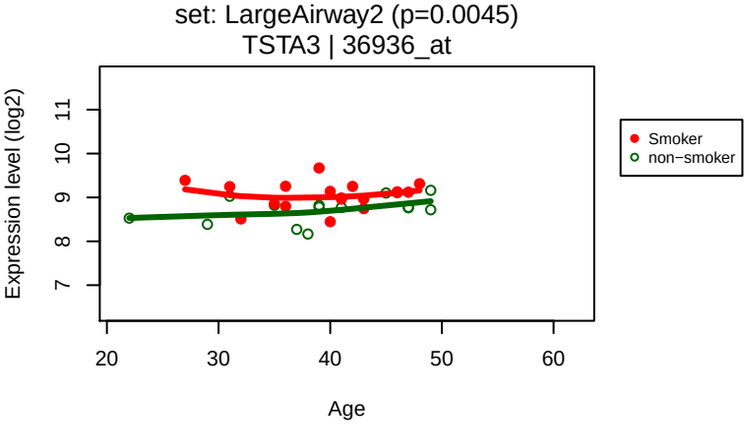

Gene expression ratio (smokers/nonsmokers)
gene: TSTA3 | 36936_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4736
p=0.4585
p=0.1001
p=0.2941
p=0.0045
p=0.031
p=0.2315
p=0.3687
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

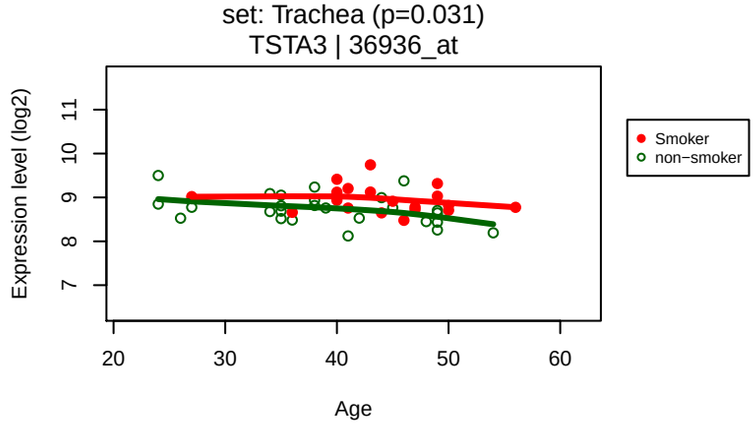

Gene expression ratio in Males
gene: TSTA3 | 36936_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6218
p=0.0196
p=0.2806
p=0.293
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TSTA3 | 36936_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1457
p=0.3654
p=0.0649
p=0.9096
SmallAirway2
LargeAirway2
Trachea
Macrophage2

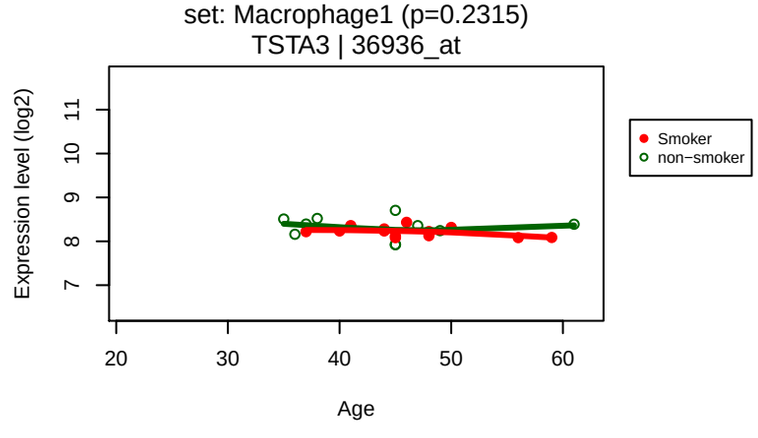

set: SmallAirway1 (p=0.4736)
TSTA3 | 36936_at
Expression level (log2)
Age
Smoker
non−smoker

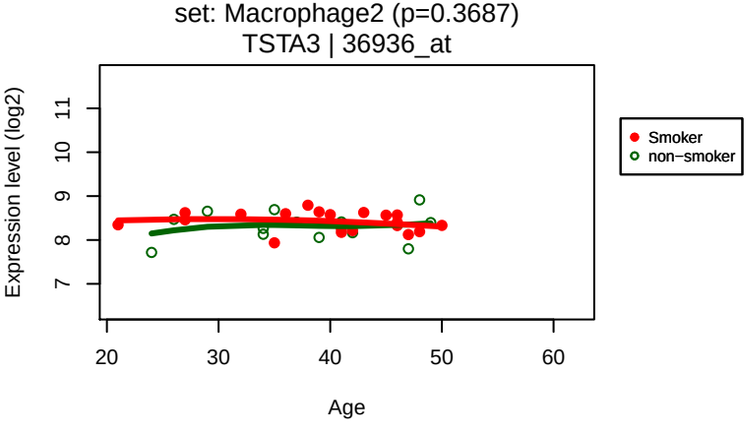

set: SmallAirway2 (p=0.4585)
TSTA3 | 36936_at
Expression level (log2)
Age
Smoker
non−smoker

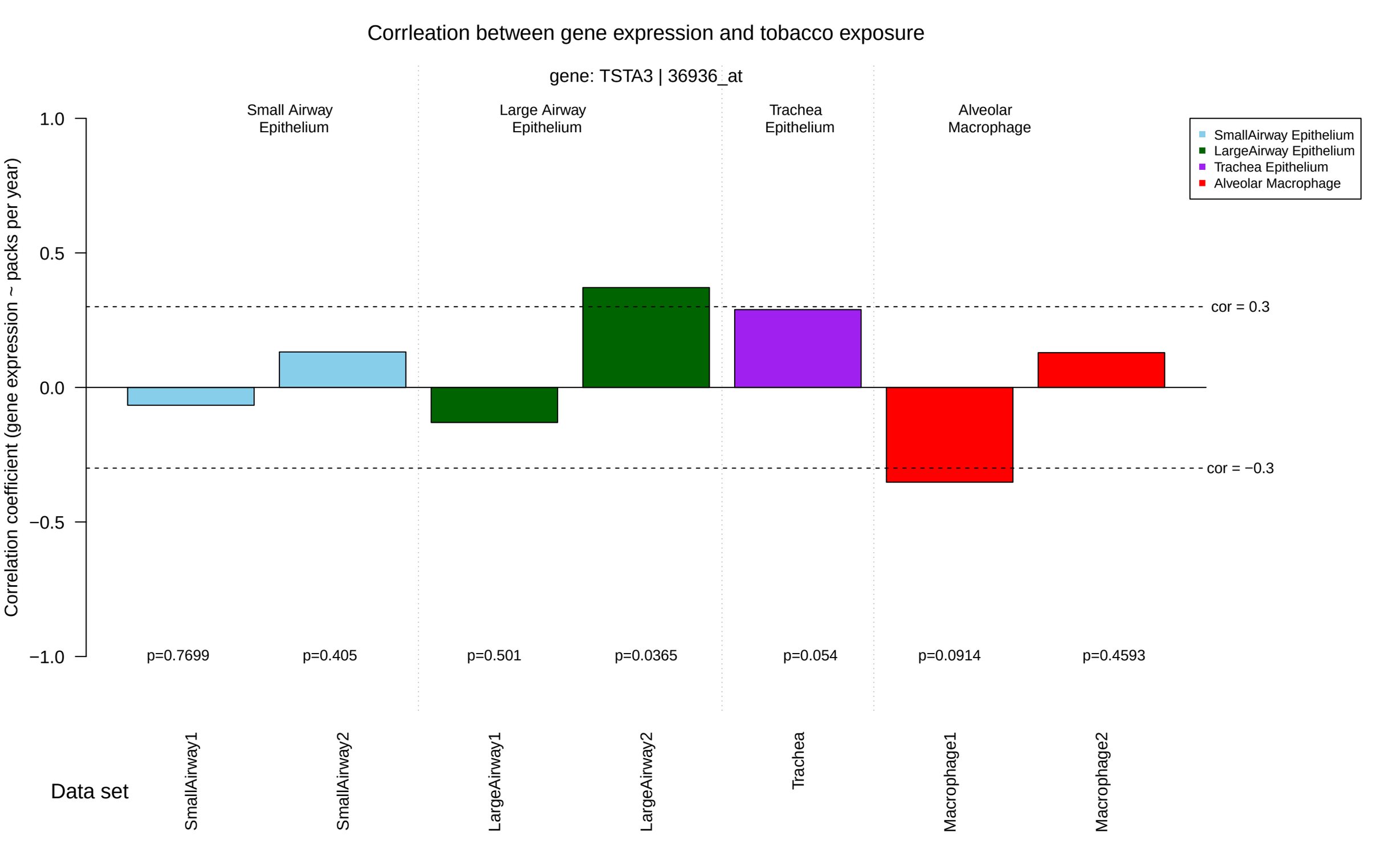

set: LargeAirway1 (p=0.2941)
TSTA3 | 36936_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0045)
TSTA3 | 36936_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.031)
TSTA3 | 36936_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2315)
TSTA3 | 36936_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3687)
TSTA3 | 36936_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TSTA3 | 36936_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7699
p=0.405
p=0.501
p=0.0365
p=0.054
p=0.0914
p=0.4593
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2